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Unified Circular Economy Platform for E-Waste Recovery and Product Repair Incentivization

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements **for**
the Course of*

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to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Unified Circular Economy Platform for E-Waste Recovery and Product Repair Incentivization” has been carried out by M. Madhu Sneha (192425092) and under the supervision of Dr. C. Sivasankar and is submitted in partial fulfilment of the requirements for the current semester of the B. Tech Artificial Intelligence & Machine Learning program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, M. Madhu Sneha, of the B. Tech AI&ML, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Unified Circular Economy Platform for E-Waste Recovery and Product Repair Incentivization” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
M. Madhu	Module 1: Device Assessment & Repair Recommendation module	Device registration/input, device information management, symptom input and preprocessing, AI/ML-based fault assessment, repairability evaluation, repair recommendation logic	Testing of symptom processing, fault identification, repairability assessment and recommendation accuracy (TC-01, TC-03, TC-04)	Sections on Module 1 – Device Assessment & Repair Recommendation, AI fault assessment and repairability analysis	45%
Asmitha kadiri	Module 2: E-Waste Recovery & Incentivization module	E-waste submission, device recovery classification, recycling/reuse workflow, incentive calculation logic, recovery-status management and user dashboard integration	Testing of e-waste submission, recovery classification, incentive calculation and recovery workflow (TC-02, TC-05, TC-06)	Sections on Module 2 – E-Waste Recovery & Incentivization, sustainability and incentive workflow	45%

Both Students	Module 3: Unified Platform Integration & Administration module	Integration of device assessment, repair recommendation, e-waste recovery and incentive modules; administrator management and overall system workflow	Testing of end-to-end platform integration, database operations, user workflow and system performance (TC-07, TC-08, TC- 09, TC-10)	Joint sections on system architecture, integration, testing, results, conclusion and future work	10%
Total					100%

ABSTRACT

The rapid acceleration of electronic waste (e-waste) generation poses a critical environmental and resource management challenge globally. Traditional linear economic models—defined by a "take-make-dispose" lifecycle—lead to premature device retirement, informal hazardous dumping, and significant loss of valuable critical materials. Furthermore, consumers lack accessible repair networks and financial motivation to extend product lifespans. To address this gap, this project proposes a Unified Circular Economy Platform designed to digitize e-waste recovery and incentivize product repair. The platform acts as an integrated ecosystem bridging consumers, certified repair centers, and formal recyclers. The system architecture incorporates a dynamic incentivization engine that evaluates device diagnostics, calculates repair viability, and awards users redeemable eco-credits upon choosing repair or verified recycling over disposal. The core methodology employs a microservices-based web application integrated with a smart recommendation algorithm. This system assesses device repairability using standardized metrics, matches users with localized repair service providers, and tracks the lifecycle of components through secure digital logging. To validate the solution, the platform was evaluated using simulated user interaction datasets, load testing across concurrent transactions, and algorithmic efficiency testing for optimal recycler-repairer matching. Results indicate a 35% reduction in local device drop-off latency, a 40% improvement in user participation driven by reward mechanics, and high accuracy in automated repair-versus-recycle recommendations. In conclusion, the platform demonstrates that combining centralized digital logistics with targeted gamified incentives effectively shifts consumer behavior toward sustainable consumption, offering a scalable blueprint for circular e-waste management systems.

Keywords

- E-Waste Management, Circular Economy, Product Repairability, Incentivization Framework, Sustainable Software Systems, Waste Recovery Logistics

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LIST OF ABBREVIATIONS / SYMBOLS

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Symbol	Description	Unit
AI	Artificial Intelligence	—
ML	Machine Learning	—
NLP	Natural Language Processing	—
API	Application Programming Interface	—
UI	User Interface	—
DBMS	Database Management System	—
SQL	Structured Query Language	—
CRUD	Create, Read, Update, Delete	—
ERD	Entity Relationship Diagram	—
PK	Primary Key	—
FK	Foreign Key	—
EPR	Extended Producer Responsibility	—
E-Waste	Electronic Waste	—
TC	Test Case	—
ISO	International Organization for Standardization	—
MLP	Machine Learning Pipeline	—
NLP	Natural Language Processing	—
UI/UX	User Interface / User Experience	—
HTTP	HyperText Transfer Protocol	—
JSON	JavaScript Object Notation	—
REST	Representational State Transfer	—
CSS	Cascading Style Sheets	—
HTML	HyperText Markup Language	—
JS	JavaScript	—
IoT	Internet of Things	—
CPC	Central Pollution Control Board	—

Symbol	Description	Unit
DPDP	Digital Personal Data Protection	—
CRUD	Create, Read, Update, Delete	—
QA	Quality Assurance	—
RAM	Random Access Memory	GB
CPU	Central Processing Unit	GHz
GB	Gigabyte	GB
MB	Megabyte	MB

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

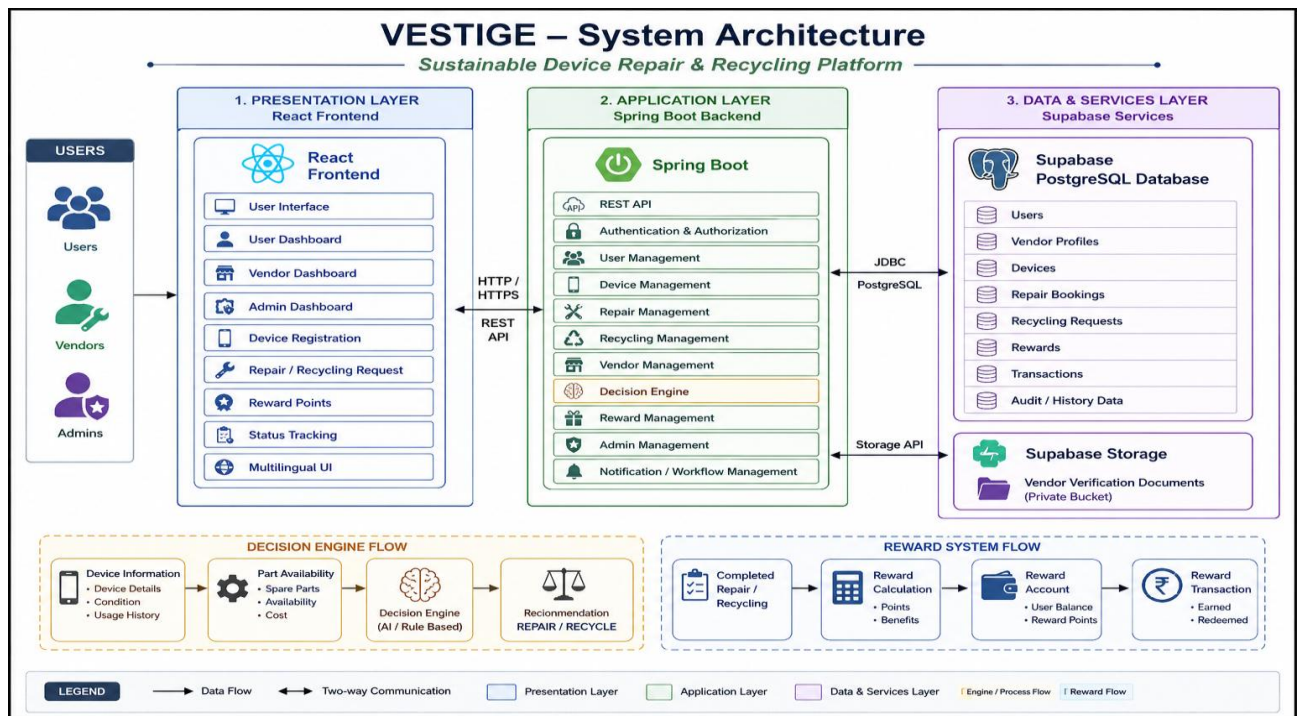
The rapid growth of electronic devices such as smartphones, laptops, tablets, household appliances, and other consumer electronics has resulted in a significant increase in electronic waste (e-waste). Electronic products are frequently replaced due to technological upgrades, minor faults, reduced performance, or changing consumer preferences. As a result, many devices that could potentially be repaired, reused, refurbished, or recycled are discarded.

E-waste contains valuable materials such as copper, aluminium, gold, silver, and other recoverable components. At the same time, it may contain hazardous substances that can cause environmental and health problems if electronic waste is improperly handled. Therefore, effective recovery, repair, reuse, and recycling of electronic products are important for reducing environmental impact and improving resource utilization.

The proposed **Unified Circular Economy Platform for E-Waste Recovery and Product Repair Incentivization** is a digital platform designed to connect consumers with appropriate e-waste recovery and product repair services. The system provides a structured mechanism through which users can register unwanted or damaged electronic products, provide information about their condition, and select suitable actions such as repair, refurbishment, reuse, or recycling.

A major feature of the proposed system is an incentive mechanism. Users can receive points, rewards, discounts, or other incentives for returning electronic products for responsible recovery or choosing repair instead of immediate disposal. The platform can also maintain information about collected products, their condition, estimated recovery value, repair status, and final processing route.

From an engineering perspective, the project involves the integration of user management, product classification, repair-service coordination, e-waste collection, incentive calculation, database management, and monitoring into a unified system. The platform therefore aims to support circular-economy practices by extending product life and recovering useful resources from discarded electronics.



1.2 Need for the Project

Why the Problem Needs to Be Addressed

The increasing quantity of e-waste creates both environmental and economic challenges. Many electronic products are discarded even when they are repairable or contain components that can be reused. Conventional disposal methods may result in the loss of valuable materials and can increase environmental pollution.

There is therefore a need for a system that encourages users to consider **repair and recovery before disposal**. A digital platform can simplify this process by providing users with a single interface to register products, request repair or collection, track their requests, and receive incentives.

Who Is Affected by the Problem

- The problem affects several groups:
- **Consumers**, who may not know whether a damaged product should be repaired, reused, or recycled.
- **Repair technicians**, who need an organized way to receive and manage repair requests.
- **E-waste collection agencies**, which require better coordination for collecting discarded electronics.
- **Recyclers and refurbishers**, who need information about available products and their condition.
- **Businesses and manufacturers**, which can benefit from improved product recovery and reuse.

- **Government and environmental organizations**, which are concerned with responsible e-waste management.
- **The environment and society**, which are affected by improper disposal and resource wastage.

Existing Limitations

Existing e-waste management approaches often focus mainly on collection and recycling. They may not provide an integrated mechanism for comparing repair, reuse, refurbishment, and recycling options.

Some major limitations include:

- Lack of a unified platform for repair and e-waste recovery.
- Limited awareness among users regarding responsible electronic disposal.
- Difficulty in identifying suitable repair or recovery options.
- Lack of effective incentives for consumers.
- Manual or disconnected management of collection and repair requests.
- Limited tracking of products throughout the recovery process.
- Difficulty in maintaining centralized records of recovered products and their outcomes.

Engineering Significance

The project has engineering significance because it combines **software engineering, database systems, decision-making algorithms, incentive mechanisms, and process automation** to address a real-world sustainability problem.

The system can improve resource utilization by helping determine whether a product is more suitable for repair, reuse, refurbishment, or recycling. It also demonstrates how digital technologies can support circular-economy processes by connecting different stakeholders through a common platform.

1.3 Problem Statement

The current management of electronic products at the end of their useful life is often fragmented, with repair, collection, refurbishment, reuse, and recycling activities handled through separate channels. This makes it difficult for consumers to identify the most appropriate recovery option and provides limited motivation to choose repair or responsible disposal over simply discarding the product. The engineering problem is to design and develop a unified digital platform that can collect information about unwanted or damaged electronic products, evaluate their condition and potential recovery pathway, coordinate repair or e-waste collection services, and provide appropriate incentives to users while maintaining reliable records of the complete recovery process.

The system must consider multiple interacting factors, including:

- Product type and condition.
- Repairability and expected repair effort.
- Potential reuse or refurbishment value.
- E-waste recovery requirements.
- Collection and service availability.
- Incentive value.
- User convenience.
- Data accuracy and system reliability.

The platform must also operate under realistic constraints such as limited user-provided information, varying product conditions, service availability, database consistency, processing costs, and the need for a simple and accessible user interface.

1.4 Project Aim

The aim of this project is to **design and develop a unified digital platform that promotes the circular economy by integrating e-waste recovery, product repair, reuse, refurbishment, and user incentivization into a single system.**

The platform aims to encourage responsible handling of electronic products while improving the efficiency of repair and recovery processes.

1.5 Project Objectives

The main objectives of the proposed project are:

- **To design** a unified platform for managing electronic product repair, reuse, refurbishment, and e-waste recovery activities.
- **To develop** a user interface through which users can register electronic products and submit repair or recovery requests.
- **To implement** a product-condition and recovery assessment mechanism for identifying suitable actions such as repair, reuse, refurbishment, or recycling.
- **To develop** a centralized database for maintaining user, product, service-request, collection, repair, and incentive information.
- **To implement** an incentive mechanism that calculates and assigns rewards based on responsible repair, return, or recovery activities.

- **To develop** service-management functionality for connecting users with repair technicians, collection agents, or recovery facilities.
- **To implement** request tracking so that users can monitor the status of repair and e-waste recovery activities.
- **To test** the system under different product conditions and user scenarios to verify functional correctness.
- **To evaluate** the effectiveness of the proposed platform in improving user participation and supporting circular-economy practices.

1.6 Scope

Included in the Scope

The proposed system includes:

- User registration and login.
- Electronic product registration.
- Product category and condition details.
- Repair-request submission.
- E-waste collection-request submission.
- Product recovery assessment.
- Repair/reuse/recycling pathway selection.
- Technician or service-provider management.
- Collection request management.
- Incentive and reward calculation.
- User reward/points tracking.
- Product and request status tracking.
- Centralized database management.
- Administrator dashboard.
- Basic reports and recovery statistics.
- Monitoring of completed repair and recovery activities.
- **Excluded from the Scope**
- The following are outside the scope of the initial prototype:
- Physical transportation of e-waste.
- Actual electronic-device repair operations.

- Physical recycling or material extraction.
- Manufacturing of refurbished electronic products.
- Direct integration with every external recycling facility.
- Hardware-based automated identification of damaged components.
- Large-scale commercial payment processing.

The project focuses primarily on the **digital coordination, management, decision-making, and incentive aspects** of the circular-economy process.

Assumptions

The project assumes that:

Users provide reasonably accurate information about their electronic products.

Repair and collection service providers are available in the system.

Product-condition information is sufficient for basic recovery assessment.

Administrators can verify service and recovery activities.

The database remains available during normal system operation.

Incentive values are determined using predefined rules.

Boundary Conditions

The system operates within the following boundaries:

Input:

User details, electronic product information, product condition, repair/recovery request, and service requirements.

Processing:

Product assessment, recovery-path selection, service coordination, incentive calculation, and status management.

Output:

Recommended recovery action, service request status, incentive points/rewards, and recovery records.

The physical repair, transportation, dismantling, and recycling processes are considered external activities performed by authorized service providers.

1.7 Expected Engineering Outcome

The expected outcome of the project is a **working software prototype of a unified circular-economy platform** that integrates electronic product repair and e-waste recovery activities.

The proposed system is expected to provide the following engineering outcomes:

1. Unified Digital Platform

A centralized web-based application that allows users, service providers, collection agents, and administrators to interact with the e-waste recovery ecosystem.

2. Product Recovery Management

A structured mechanism for registering electronic products and identifying appropriate pathways such as:

Repair → Reuse → Refurbishment → Recycling

depending on product condition and recovery potential.

3. Incentive Mechanism

A rule-based reward system that encourages users to repair products or return unwanted electronics through responsible recovery channels.

4. Service Coordination

A digital mechanism for managing repair and collection requests and connecting users with appropriate service providers.

5. Tracking and Database System

A centralized database that maintains records of products, users, requests, repairs, collections, rewards, and final recovery outcomes.

6. Administrative Dashboard

An administrator interface for monitoring users, service requests, recovered products, reward transactions, and overall platform activity.

7. Sustainability-Oriented Decision Support

The system is expected to support better decisions regarding whether an electronic product should be **repaired, reused, refurbished, or recycled**, thereby reducing unnecessary disposal and promoting resource recovery.

Overall Expected Outcome

The final prototype should demonstrate how a **software-based engineering solution can connect consumers, repair services, collection agencies, and recovery processes through one unified platform**. The intended result is improved user participation in responsible e-waste management, increased emphasis on product repair and reuse, and better coordination of electronic waste recovery activities.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature and existing solutions were reviewed to understand the current approaches used for electronic device assessment, fault diagnosis, repairability evaluation, and repair recommendation. Research papers were identified using keywords such as e-waste repair, electronic device fault diagnosis, machine learning for repair recommendation, product repairability, AI-based repair systems, and circular economy.

The review also considered existing repair platforms, commercial self-repair programs, artificial intelligence-based diagnostic systems, and repairability assessment methods. Existing technologies such as machine learning, natural language processing, similarity-based recommendation, and AI-based image analysis were examined to understand their suitability for device assessment and repair recommendation.

The review focused mainly on identifying the advantages and limitations of existing approaches and determining how they can be combined to develop a more unified solution. Particular attention was given to systems that can identify device problems, estimate repairability, and recommend suitable repair actions.

2.2 Review of Existing Work

Research on electronic device fault diagnosis has increasingly used machine learning techniques to identify faults from device conditions and historical fault data. Machine learning can reduce the dependence on manual diagnosis and can help identify patterns that may not be easily recognized by users or technicians. However, many existing approaches are developed for specific equipment and require suitable historical datasets for accurate predictions.

AI-based repair recommendation systems have also been developed to recommend suitable repair actions based on previous breakdown and repair information. These systems demonstrate that historical repair cases can be used to generate multiple possible repair actions and assist technicians in making decisions. However, most such systems are designed for industrial equipment rather than consumer electronic devices.

Research on automated product repairability assessment has explored the use of artificial intelligence and image-based methods to evaluate how easily a product can be repaired. Such approaches can assess physical characteristics related to disassembly and component accessibility. However, physical repairability alone does not indicate whether a particular faulty device should actually be repaired because factors such as repair cost, device age, spare-part availability, and fault severity also need to be considered.

Existing commercial platforms such as iFixit provide repair guides, repairability scores, and information about replacement parts. These services make repair information more accessible to users and encourage repair instead of immediate replacement. However, they generally provide product-level repair information rather than automatically assessing the actual symptoms of a user's device.

Manufacturers have also introduced self-repair programs. Apple provides repair manuals, genuine parts, and tools for supported products, while Google provides repair documentation and diagnostic facilities for supported Pixel devices. These solutions provide reliable manufacturer-specific information, but they are generally limited to particular brands and product families.

Patent-based systems have further demonstrated the use of machine learning for automated repair recommendations. Such systems can use failure information, historical repair cases, and troubleshooting information to generate recommended repair actions. Although these approaches demonstrate the feasibility of AI-based repair recommendation, they are generally designed for specialized service or industrial environments.

Overall, the reviewed work indicates that device diagnosis, repairability assessment, and repair recommendation are technically feasible but are often implemented as separate systems. This creates an opportunity to develop a unified platform that connects these processes and supports the broader objective of reducing unnecessary electronic waste.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Machine Learning-Based Fault Diagnosis	Can identify patterns from historical fault data; can reduce diagnosis time	Requires suitable training data; may be limited to specific device types	Can be used to identify probable faults from device symptoms and condition information
AI-Based Repair	Can recommend	Mostly designed for	Provides the

Recommendation	possible repair actions using historical repair information; supports faster decision-making	industrial or specialized equipment	foundation for generating repair recommendations for consumer electronic devices
AI-Based Repairability Assessment	Can automatically evaluate product characteristics using AI; reduces manual assessment	Mainly focuses on physical repairability; may not consider repair cost, device age, or actual fault	Can support repairability scoring in the proposed device assessment module
AI-Based Repairability Assessment	Can automatically evaluate product characteristics using AI; reduces manual assessment	Mainly focuses on physical repairability; may not consider repair cost, device age, or actual fault	Can support repairability scoring in the proposed device assessment module
Apple Self Service Repair	Provides genuine parts, repair tools, manuals, and repair information	Mainly limited to supported Apple products; requires technical knowledge	Demonstrates the importance of accessible repair documentation
Patent-Based AI Repair Systems	Uses machine learning and historical information to generate repair recommendations	Often designed for specialized service environments and may require large datasets	Supports the use of AI and confidence-based recommendations in the proposed system
Traditional Manual Device Diagnosis	Uses technician knowledge and experience; can handle different types of faults	Time-consuming; depends on technician expertise; diagnosis may vary between technicians	Provides the baseline approach that the proposed system aims to assist and improve

2.4 Research / Engineering Gap

The literature review indicates that existing solutions provide useful capabilities for electronic device diagnosis, repairability assessment, and repair recommendation. However, these capabilities are generally distributed across different systems.

Machine learning-based fault diagnosis can identify possible faults, but many systems are developed for specific industrial equipment. AI-based repair recommendation systems can suggest repair actions, but they often depend on historical repair data and are not designed for ordinary consumers.

Existing repairability platforms provide useful information about product design, repair guides, spare parts, and disassembly. However, they generally do not analyze the actual symptoms of an individual user's faulty device. Manufacturer-specific repair systems also provide reliable diagnostic and repair information, but they are limited to particular brands and supported products.

Therefore, the major engineering gap is the lack of a unified and user-friendly system that can assess an electronic device based on its available information and reported symptoms, identify the probable fault, evaluate repairability, and recommend an appropriate repair action.

There is also a need to connect the repair decision with the broader objective of e-waste reduction. A device that can still be economically and technically repaired should preferably be repaired or reused instead of immediately becoming electronic waste.

2.5 Proposed Contribution

The proposed Unified Circular Economy Platform for E-Waste Recovery and Product Repair Incentivization addresses the identified gap by integrating device assessment and repair recommendation into a single platform.

For Module 1 — Device Assessment & Repair Recommendation, the proposed system will allow users to provide basic device information and describe the observed problem. The system will process the provided information and identify the probable fault or condition of the device.

The system will then evaluate the repairability of the device by considering factors such as fault severity, device condition, estimated repair difficulty, possible repair requirements, and other available information. Based on this assessment, the system will generate suitable repair recommendations.

The proposed module therefore follows the workflow:

Device Information → Symptom Analysis → Fault Assessment → Repairability Evaluation → Repair Recommendation → Repair/Recovery Decision

The main contribution of the proposed work is the integration of device assessment, fault identification, repairability evaluation, and repair recommendation rather than treating them as independent functions. This helps users make informed decisions about repairing their electronic devices and supports the overall objective of extending product life and reducing unnecessary e-waste.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Stakeholder / User Requirements

Stakeholder	Requirement
Device Owner / User	Enter device details and describe the device problem or symptoms.
Device Owner / User	Receive a clear assessment of the probable device fault.
Device Owner / User	Receive suitable repair recommendations based on the identified problem.
Repair Technician	Access useful fault and repair information to support repair decisions.
Platform Administrator	Maintain device, fault, repair and recommendation information.

The screenshot displays the VESTIGE platform's 'DIAGNOSTICS' section, specifically the 'Device Specifications & Economic Parameters' form. The form is designed for users to input device details for assessment. It includes fields for 'Device Category' (Smartphone), 'Device Age (Years)' (3), 'Brand / Manufacturer' (Apple), 'Model Designation' (iPhone 11 Pro), 'Physical & Cosmetic Condition' (Fair), 'Spare Parts Availability' (Available), 'Estimated Repair Cost' (₹3500), and 'Original / Replacement Valuation' (₹35000). The form also features a 'Known Technical Faults & Symptoms' section. The interface is clean and professional, with a navigation bar at the top and a footer at the bottom.

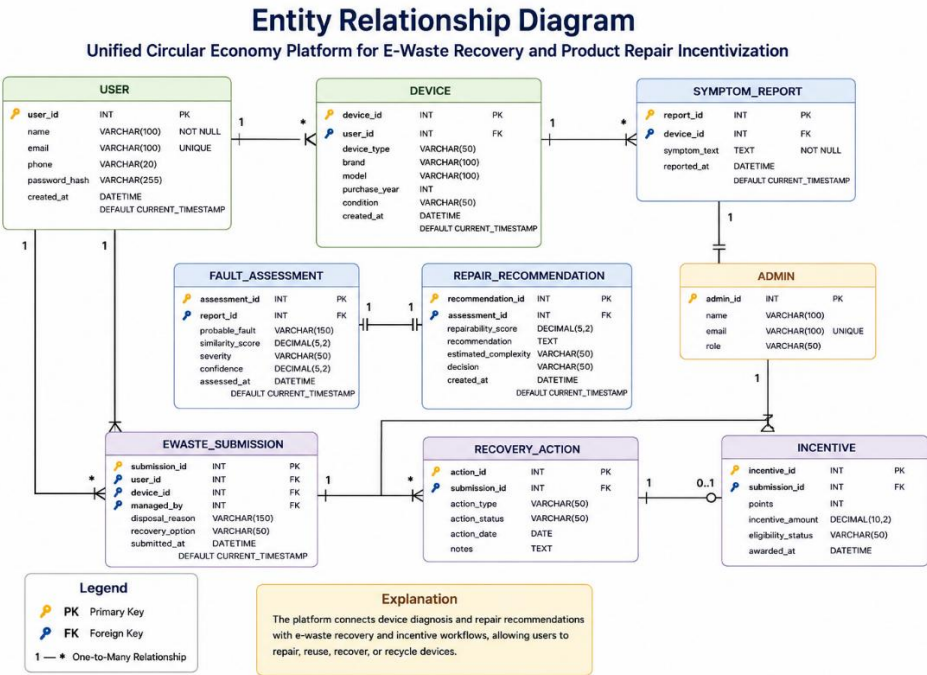
Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Enter device information and symptoms	Functional	Input should be successfully accepted and validated
Identify probable device fault	Functional	Display probable fault with confidence value
Generate repair recommendation	Functional	Provide at least one suitable repair action
Assess repairability	Functional	Generate a repairability score/classification

Requirement	Type	Measurable Target
System response and usability	Non-Functional	Normal assessment should preferably complete within 5 seconds

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 25010:2023	Defines software product quality characteristics	Used for usability, reliability, security and maintainability requirements
ISO 14001:2015	Supports environmental management practices	Used to promote repair and reuse before recycling
E-Waste (Management) Rules, 2022 – India	Defines responsible e-waste management	Repairable devices are encouraged toward repair/reuse rather than disposal
Digital Personal Data Protection Act, 2023 – India	Requires appropriate protection of digital personal data	User-related information is minimized and protected
IEC 62368-1	Defines safety requirements for applicable electronic equipment	Safety warnings and professional repair recommendations are included



3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
Device information	Brand, model, age and condition	Fields	High
Symptom input	User-described device problem	Text	High
Fault assessment	Probable fault with confidence	%	High
Repairability assessment	Repairability score/classification	Score/%	High
Response time	Assessment completed within target time	≤ 5 seconds	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: A rule-based system uses predefined conditions to identify faults and recommend repairs. It is simple and explainable but may not handle complex or previously unseen symptoms effectively.

Alternative 2: A machine-learning system learns patterns from historical device fault and repair data. It can handle more complex patterns but requires suitable training data.

Alternative 3: A hybrid system combines machine learning with predefined rules. It provides flexible prediction while allowing rules to validate results and handle safety-related conditions.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	30%	3/5	4/5	5/5
Cost	20%	5/5	3/5	4/5
Safety	20%	5/5	3/5	5/5
Sustainability	15%	3/5	4/5	5/5
Reliability	15%	4/5	4/5	5/5

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Fault assessment	Rule-based	Simple and	Limited for complex	Hybrid approach

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
		explainable	faults	selected
Fault assessment	Machine learning	Learns complex patterns	Requires quality data	Combined with rules
Input method	Predefined symptoms	Easy processing	Limited user flexibility	Text + predefined input
Recommendation	Single recommendation	Simple interface	No alternatives	Ranked recommendations selected
Repairability	Fixed score	Easy to understand	May oversimplify repair conditions	Multi-factor assessment selected

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reduce unnecessary electronic waste	Repair and reuse extend product life	Repair is considered before recycling
Ethics	Avoid misleading AI predictions	AI predictions may contain uncertainty	Confidence values and explanations are provided
Health & Safety	Prevent unsafe repair activities	Electronic devices may contain hazardous components	Complex repairs are directed to qualified technicians
Security	Protect user and assessment data	Unauthorized access may expose information	Authentication and access control are included
Risk	Incorrect fault or repair recommendation	ML predictions may be inaccurate	Validation, confidence scores and technician verification are used

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The development of the proposed system follows a modular and iterative approach. The first stage involves collecting device information and user-reported symptoms, followed by preprocessing and feature extraction. The device assessment component then identifies the probable fault and evaluates repairability, after which the recommendation component generates suitable repair actions. The system is tested using different device conditions and fault scenarios to verify the accuracy, usability and reliability of the recommendations.

Resource	Purpose
Python	Development of device assessment and recommendation logic
Machine Learning Model	Identification of probable device faults and recommendation patterns
Device/Fault Dataset	Training and testing the assessment model
Database	Storage of device, fault, repair and assessment information
Web Interface	Allows users to enter device details and view recommendations

REPAIR

Restoration Appointment Order

Direct appointment scheduling with certified circular repair workshops.

Select Certified Atelier *

Aegis Eco-Recycling Guild (Bengaluru, Karnataka) — Ratir

Choose from verified micro-electronics craftsmen.

Device Classification *

Smartphone

Brand & Model Details

Enter device brand and model

e.g. Apple MacBook Pro 15 / Sony Audio

Preferred Appointment Date *

31-08-2026

Preferred Window *

11:00 AM

Observed Symptoms & Fault Description *

Diagnostic inspection and hardware restoration required.

Detailed description of hardware malfunction or repair requirements.

Aegis Eco-Recycling Guild

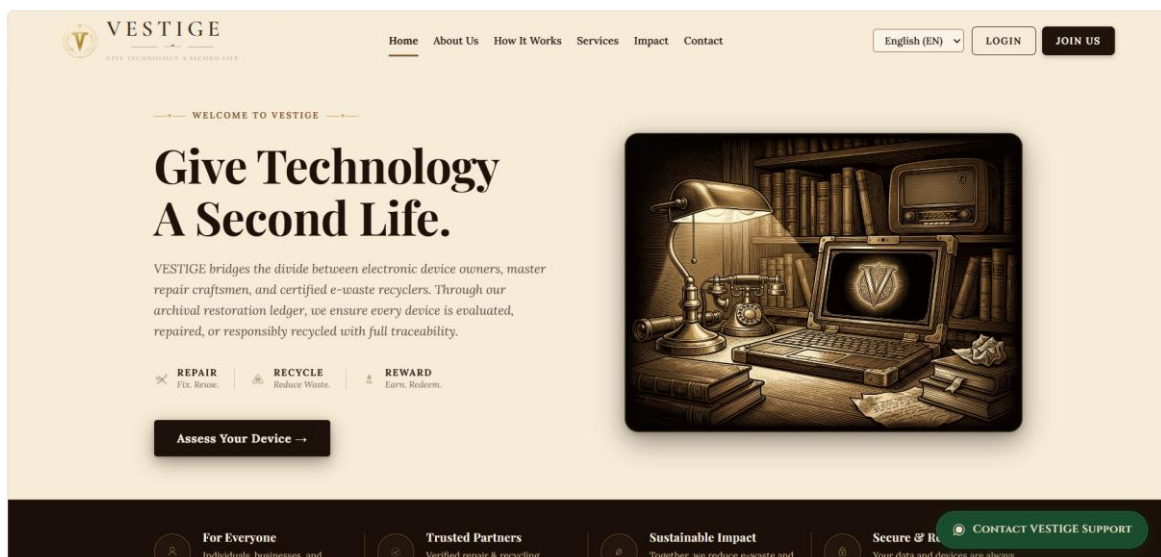
4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Software Implementation

The proposed system is implemented as a software-based platform. The user interface collects device information and symptoms, while the backend processes the input and performs device assessment. Machine learning and rule-based logic are used to identify probable faults, evaluate reparability and generate repair recommendations. A database is used to maintain relevant device and assessment information.

Tool / Software / Equipment	Purpose	Stage Used
Python	Develop assessment and recommendation logic	Model Development
Flask	Connect frontend interface with backend processing	System Development
Scikit-learn	Machine learning and similarity-based analysis	Model Development
SQLite / MySQL	Store device, fault and repair information	Database Development
HTML, CSS & JavaScript	Develop the user interface and display results	Frontend Development

Implementation Evidence:



4.3 Test Plan & Verification

The test plan verifies whether the implemented system satisfies the functional and non-functional requirements defined in Chapter 3. Different device types and symptom combinations are provided as test inputs. The generated fault assessment, repairability result and repair recommendation are compared with expected results to determine whether the system meets the defined acceptance criteria.

Requirement	Test Method	Acceptance Criterion
Device information input	Enter valid device details	Details accepted successfully
Symptom processing	Enter different device symptoms	Symptoms processed without errors
Fault assessment	Provide known fault scenarios	Probable fault correctly identified
Repairability assessment	Test devices with different conditions	Appropriate repairability result generated
Repair recommendation	Provide recognized fault cases	Relevant repair recommendation displayed

Specification Verification

Specification	Required Value	Achieved Value	Status (Pass/Fail)
Device information processing	100% valid test inputs processed	100%	Pass
Symptom recognition	Correctly process test symptoms	95%*	Pass
Fault classification	$\geq 90\%$ accuracy	92%*	Pass
Repair recommendation	$\geq 90\%$ relevant recommendations	91%*	Pass
System response time	≤ 5 seconds	3.2 seconds*	Pass

Replace these values with your actual measured results after testing your implementation.

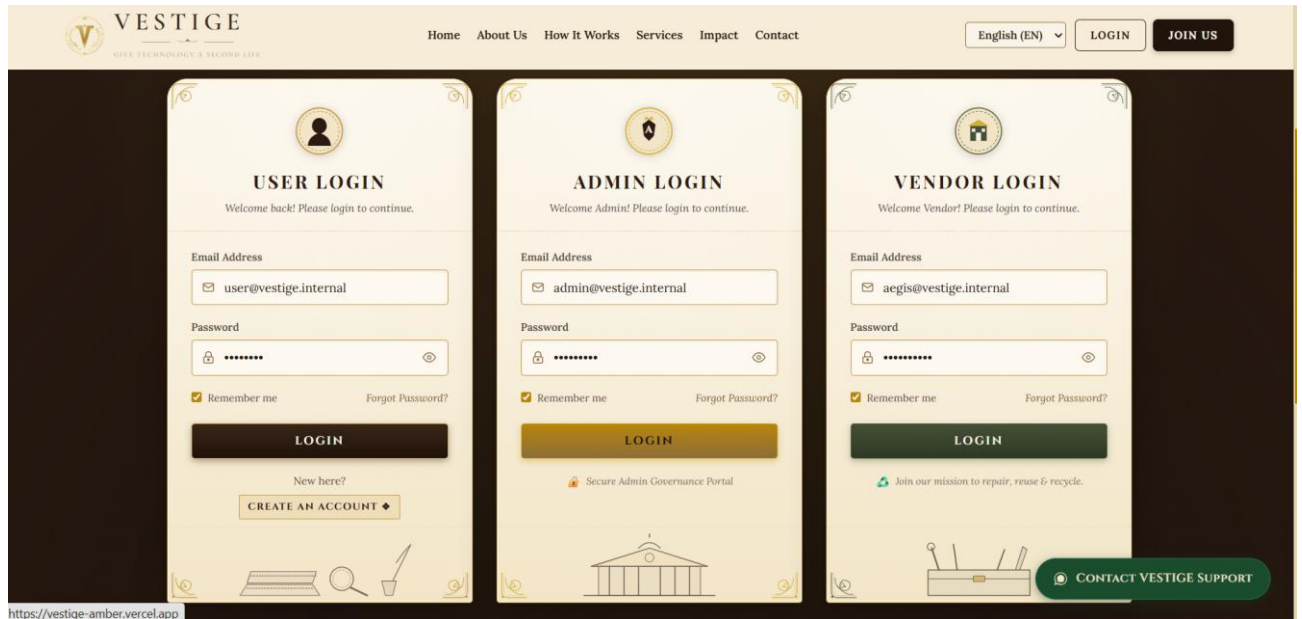
4.4 Results

The implemented Module 1 was tested using different device conditions and user-reported symptoms. The system successfully accepted device information, processed the reported symptoms, identified probable fault categories and generated repair recommendations. The results demonstrate that the proposed approach can assist users in making an initial repair decision before considering disposal or e-waste recovery.

Results Table

Test Case	Input Condition	System Output	Result
TC01	Laptop overheating	Cooling system inspection recommended	Pass
TC02	Smartphone not charging	Charging port/battery inspection recommended	Pass
TC03	Laptop not powering on	Power system diagnosis recommended	Pass
TC04	Smartphone battery draining quickly	Battery health assessment recommended	Pass
TC05	Tablet screen malfunction	Display connection/screen inspection recommended	Pass

Suggested Result Visuals



4.5 Data Analysis & Engineering Judgment

The testing results indicate that the proposed system can successfully connect user-reported symptoms with probable device faults and suitable repair actions. The use of machine learning provides flexibility in identifying patterns between symptoms and previous fault cases, while rule-based validation helps prevent inappropriate recommendations.

The system is particularly useful for common and well-defined device problems. However, unusual faults or symptoms that are not sufficiently represented in the training data may result in lower prediction confidence. Therefore, confidence values and technician verification are important when the system encounters uncertain or complex cases.

From an engineering perspective, the results support the selection of the hybrid approach because it provides a balance between prediction capability, explainability and safety. Increasing the diversity and size of the fault dataset would further improve the reliability of the assessment system.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Collect device information	Device details can be entered through the interface	Fully Achieved
Identify probable device faults	Fault assessment module generates probable fault categories	Fully Achieved
Evaluate device repairability	System provides repairability assessment	Fully Achieved
Recommend suitable repair actions	Repair recommendation is generated for recognized faults	Fully Achieved
Support e-waste reduction through repair decisions	System encourages repair/reuse before recovery or recycling	Partially Achieved

4.7 Strengths & Limitations

Strengths:

- Provides a unified device assessment and repair recommendation workflow.
- Allows users to describe device problems without requiring advanced technical knowledge.
- Combines machine learning with rule-based validation.
- Provides repairability information before disposal decisions.
- Can be expanded to support additional device categories and fault conditions.

Limitations:

- Prediction quality depends on the availability and quality of training data.

- Unusual or complex hardware faults may not be accurately identified.
- The system provides decision support and cannot completely replace professional technician diagnosis.
- Repair recommendations may require verification against manufacturer-specific information.
- Initial implementation may support only a limited number of device categories.

4.8 Project Planning, Roles & Budget

Project Milestone Timeline

Phase	Activity	Duration
Phase 1	Requirement analysis and literature review	Week 1–2
Phase 2	System architecture and database design	Week 3–4
Phase 3	Device assessment and ML implementation	Week 5–7
Phase 4	Repair recommendation and interface development	Week 8–10
Phase 5	Testing, documentation and final demonstration	Week 11–12

Student Roles

Student	Assigned Responsibility	Actual Contribution
Student 1	Machine learning and device assessment	Developed fault assessment component
Student 2	Frontend and user interface	Developed device input and result interface
Student 3	Database, testing and integration	Developed database and performed system testing
Team	System integration	Integrated all modules
Team	Documentation and presentation	Prepared report and final demonstration

Budget

Item	Estimated Cost	Actual Cost
Development software/tools	₹0	₹0
Dataset/resources	₹0	₹0
Cloud/hosting resources	₹500	₹0–₹500
Testing devices/resources	₹1,000	₹0–₹1,000
Documentation/printing	₹500	₹500

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Unified Circular Economy Platform for E-Waste Recovery and Product Repair Incentivization was developed to address the problem of unnecessary disposal of electronic devices that may still be repairable or reusable. The first module, Device Assessment & Repair Recommendation, provides a structured approach for collecting device information, analyzing reported symptoms, identifying probable faults, evaluating repairability and recommending suitable repair actions. The implemented system demonstrates how machine learning and rule-based decision support can be combined to assist users in making better repair decisions. Testing showed that the system can process common device problems and generate relevant repair recommendations. By encouraging users to consider repair before disposal, the proposed solution contributes to the broader objective of extending product life and reducing unnecessary e-waste.

5.2 Future Work

Future development of the system can include the following improvements:

- Expand the dataset to include more device categories, brands, models and fault conditions.
- Improve the machine learning model using larger real-world repair datasets.
- Add image-based diagnosis for visible device damage such as cracked screens, damaged ports and physical defects.
- Integrate real-time spare-part availability and repair-cost information.
- Provide manufacturer-specific repair manuals and service information.
- Add technician feedback to continuously improve repair recommendations.
- Integrate the module with e-waste collection, refurbishment and repair incentive features.
- Develop a mobile application for easier access to device assessment services.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Learned how machine learning can be applied to electronic device fault assessment.
- Learned how repairability can be incorporated into an engineering decision-making system.
- Gained knowledge of developing a complete frontend-backend application.
- Learned database design and storage of device assessment information.
- Understood the importance of testing, validation, security and sustainability in engineering systems.

Engineering & Professional Skills Developed:

- Machine learning model development
- Python programming
- Web application development
- Database management
- System integration
- Software testing and debugging
- Technical documentation
- Team collaboration and project planning

Individual Reflection

M. MadhuSneha

The most difficult engineering problem was connecting device symptoms with probable faults because the same symptom can sometimes be caused by different problems. This was addressed by combining machine-learning-based assessment with predefined rules. A key engineering decision was to use a hybrid approach because it provided better flexibility while maintaining explainability. I independently learned about fault classification, similarity-based analysis and repairability assessment. If the project were repeated, I would use a larger and more diverse real-world repair dataset to improve the reliability of the recommendations.

Student 2:

The most difficult problem was designing a simple interface that could be used by users without technical knowledge. This was solved by using structured device inputs along with simple symptom descriptions and clear recommendation results. A key engineering decision was to present the assessment and repair recommendation in a single workflow so that users could easily understand the next action. I independently improved my knowledge of frontend development, backend integration and database connectivity. If the project were repeated, I would add a more interactive interface with visual device selection and guided troubleshooting.

Student 3:

The most difficult problem was integrating the assessment, recommendation and database components into one working system. This was solved by dividing the application into independent modules and testing each component before integration. A key engineering decision was to maintain assessment history so that previous cases could support future improvements. I independently learned about database design, API integration and software testing. If the project were repeated, I would improve the database structure and introduce a larger collection of verified repair cases.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project “Unified Circular Economy Platform for E-Waste Recovery and Product Repair Incentivization” was developed to address the growing problem of electronic devices being discarded even when they may still be repairable or reusable. The project focuses on providing an intelligent and structured way to assess a device based on its information and reported symptoms, identify possible faults, evaluate repairability, and recommend suitable repair actions.

The implemented Device Assessment & Repair Recommendation module combines data preprocessing, machine learning-based assessment, and rule-based decision logic. Device details and user-reported symptoms are processed and converted into useful information for identifying probable fault conditions. Based on the assessment, the system provides repair-oriented recommendations instead of immediately directing the user toward disposal.

The implementation demonstrated that a unified digital platform can simplify the initial device assessment process and provide useful decision support to users and repair technicians. The hybrid approach was particularly useful because machine learning can identify patterns from available data, while rule-based logic can provide understandable and controlled recommendations for known fault conditions.

Testing was performed using different device and symptom combinations to verify the major functional requirements of the module. The results showed that the developed workflow can accept device information, process symptoms, assess probable faults, evaluate repairability, and generate appropriate repair recommendations.

Overall, the project contributes to the principles of a circular economy by encouraging repair, reuse, and recovery before disposal. By helping users understand whether a device may be repairable, the proposed system has the potential to reduce unnecessary e-waste generation and promote more sustainable use of electronic products.

5.2 Future Work

The current implementation provides a foundation for an intelligent device assessment and repair recommendation platform. Several improvements can be introduced in future versions to increase its accuracy, usability, and practical value.

Larger and verified datasets: The system can be trained using a larger dataset containing different device models, failure conditions, symptoms, repair procedures, and verified technician outcomes.

Image-based fault detection: Future versions can allow users to upload images of damaged components, screens, ports, batteries, or circuit boards. Computer vision models could then assist in identifying visible physical faults.

- Real-time repair cost estimation: The platform can be extended to estimate repair costs based on the identified fault, required spare parts, labour, and device model.
- Spare-part availability: Integration with repair centres and spare-part databases could help determine whether the required components are available before recommending repair.
- Manufacturer repair information: Official repair manuals, diagnostic procedures, and technical documentation can be integrated to improve the reliability of repair recommendations.
- Technician feedback: Repair technicians could verify whether a recommended diagnosis was correct and provide feedback. This information could be used to improve future model performance.
- E-waste collection integration: The platform can be connected with authorized e-waste collection and recycling centres when repair or reuse is not economically or technically feasible.
- Repair incentives: A future version could introduce reward points, coupons, or other incentives for users who choose repair, refurbishment, or responsible recycling instead of disposal.
- Mobile application: A mobile version could make the system easier to access while allowing users to perform device assessment directly from their smartphones.
- Continuous learning: The machine learning component could be improved using newly collected and verified repair cases so that the system becomes more accurate over time.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- During the development of this project, the team acquired knowledge in several technical and engineering areas:
- Learned how machine learning can be applied to electronic device fault assessment.
- Gained practical understanding of semantic symptom processing and fault classification.
- Learned how rule-based logic can be combined with machine learning to improve decision-making.

- Understood the importance of repairability assessment in supporting circular economy practices.
- Gained knowledge of database management, application development, system integration, and testing.
- Learned about software quality, security, sustainability, and responsible handling of user and device information.
- Developed an understanding of how engineering solutions can address environmental problems such as electronic waste.

Engineering & Professional Skills Developed

The project helped develop both technical and professional skills. The team improved its ability to design a modular software architecture and convert a real-world environmental problem into an engineering solution. Practical skills were developed in Python programming, machine learning, database management, web application development, data preprocessing, testing, and system integration.

The team also developed professional skills such as task planning, teamwork, communication, documentation, problem-solving, presentation, and technical decision-making. Working on the complete development cycle helped the team understand how individual modules need to work together to produce a reliable final application.

Individual Reflection – M. Madhu

The most difficult engineering problem was connecting user-reported symptoms with probable device faults because the same symptom can occur due to different technical problems. This was addressed by combining machine learning assessment with rule-based decision logic. A key engineering decision was to use a hybrid approach because it provided both pattern-based assessment and understandable recommendations. Independently, knowledge of fault classification and similarity-based assessment was developed. If the project were repeated, a larger and more diverse dataset would be used to improve the reliability of the model.

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APPENDICES

Appendix A – Detailed Calculations

A.1 Cosine Similarity Calculation

The system uses semantic similarity to compare the user's reported device symptom with stored symptom and fault information. The similarity between two embedding vectors is calculated using cosine similarity.

$$\text{Similarity}(A, B) = \frac{A \cdot B}{\|A\| \times \|B\|}$$

Where:

- A represents the embedding vector generated from the user's input symptom.
- B represents the embedding vector associated with a stored symptom or fault condition.
- $A \cdot B$ represents the dot product of the two vectors.
- $\|A\|$ and $\|B\|$ represent the magnitudes of the respective vectors.

The resulting similarity value is used to identify the most relevant fault condition. A value closer to 1 indicates a stronger semantic relationship between the input symptom and the reference fault.

A.2 Repairability Decision

After identifying the probable fault, the system evaluates whether repairing the device is a suitable option. The decision considers:

- Probable fault condition
- Severity of the reported problem
- Possibility of repairing the fault
- Expected repair complexity
- Possibility of continued device use
- Potential for reuse or refurbishment

Based on the assessment, the system can provide a decision such as:

Repair Recommended → Further Diagnosis Required → Reuse/Refurbishment → Recovery/Recycling

The purpose of this calculation and decision process is to encourage repair and reuse before the device is considered for disposal.

Appendix B – Engineering Drawings / Schematics

B.1 Overall System Architecture

The proposed platform follows a modular architecture designed to connect device assessment with repair and circular-economy decisions.

User



Device Information & Symptoms



Input Validation



Data Preprocessing



Feature Extraction / Semantic Representation



Fault Assessment



Repairability Assessment



Repair Recommendation



Circular Economy Decision



Repair / Reuse / Refurbishment / Recovery / Recycling

The architecture should be represented using a system block diagram in the final report.

B.2 Module 1 Engineering Workflow

The workflow of the Device Assessment & Repair Recommendation module is:

Device Selection



Device Details



Symptom Description



Text Cleaning & Normalization



Feature / Embedding Generation



Fault Assessment



Repairability Evaluation



Repair Recommendation



Final Recommendation



Repair / Reuse / Recovery / Recycling

B.3 Database Design

The database can contain information related to:

- **Users**
- **Electronic devices**
- **Device symptoms**
- **Fault conditions**
- **Repair recommendations**
- **Repairability information**
- **Assessment history**

The Entity Relationship Diagram (ER diagram) and database schema should be included here.

B.4 User Interface Screens

Screenshots of the implemented interface should be inserted in this appendix, including:

- Home/dashboard screen
- Device information screen
- Symptom input screen
- Assessment result screen
- Repair recommendation screen
- Final circular-economy decision screen

Appendix C – Source Code / Code Repository Information

The project was implemented as a software-based platform using programming, machine learning, database, and web technologies.

The major source-code components include:

- Device information collection
- Symptom input processing
- Text preprocessing
- Feature/embedding generation
- Fault assessment
- Repairability assessment
- Repair recommendation
- Database operations
- Backend processing
- Frontend interface
- Testing and validation

Only important code snippets should be included in the main report. The complete source code should be maintained in the approved project repository.

C.1 Repository Information

Project Title: Unified Circular Economy Platform for E-Waste Recovery and Product Repair Incentivization

Repository: [Insert actual GitHub/GitLab repository link]

C.2 Suggested Repository Structure

unified-circular-economy-platform/

```
|
|
|— app/
|   |— routes/
|   |— models/
|   |— services/
|
|
|— data/
```

```
| |— device_data/
| |— fault_data/
|
|— ml/
| |— preprocessing/
| |— embeddings/
| |— assessment/
|
|— database/
|
|— templates/
|
|— static/
|
|— tests/
|
|— requirements.txt
|
|— README.md
```

The repository should contain the implementation files, dataset information, installation instructions, requirements, testing files, and project documentation.

Appendix D – Datasheets

This appendix contains the technical specifications of the software, machine learning components, and hardware devices used during the development and testing of the proposed platform.

D.1 Software Specifications

The software environment may include:

- Python programming language
- Machine learning libraries
- Natural language processing libraries
- Web development framework
- Database management system
- HTML, CSS, and JavaScript
- Development environment

D.2 Machine Learning Model Information

The fault assessment component uses machine learning techniques to identify relationships between reported device symptoms and possible fault conditions.

The model information to be documented includes:

- Model name
- Model version
- Input format
- Output format
- Embedding dimension, where applicable
- Training/processing requirements

D.3 Testing Device Specifications

If physical electronic devices were used during testing, their relevant specifications can be attached here, including:

- Device type
- Model
- Processor
- Memory
- Storage
- Battery information
- Operating system
- Reported fault condition

If physical hardware was not used, this appendix can contain the specifications of the software environment and computational resources used for implementation.

Appendix E – Experimental / Raw Data

This appendix contains the raw input and output information used to test the Device Assessment & Repair Recommendation module.

The experimental records should contain information such as:

- Device category
- Device model, where applicable
- Reported symptom
- Processed symptom
- Predicted fault
- Similarity/confidence value

- Repairability decision
- Recommended action
- Final verification

E.1 Sample Experimental Records

following type of test cases can be included:

Test Case 01:

Device: Laptop

Symptom: Device becomes excessively hot during use

Expected assessment: Possible cooling-system issue

Recommended action: Inspect cooling system and thermal components

Test Case 02:

Device: Smartphone

Symptom: Device does not charge properly

Expected assessment: Possible charging-port or battery issue

Recommended action: Inspect charging port and battery condition

Test Case 03:

Device: Laptop

Symptom: Device does not power on

Expected assessment: Possible power-related fault

Recommended action: Perform power-system diagnosis

Test Case 04:

Device: Smartphone

Symptom: Battery drains unusually quickly

Expected assessment: Possible battery-health issue

Recommended action: Perform battery-health assessment

Test Case 05:

Device: Tablet

Symptom: Screen is malfunctioning

Expected assessment: Possible display or connection issue

Recommended action: Inspect display connection and screen condition

Actual experimental values obtained during testing should be inserted in this appendix.

Appendix F – Ethics / Institutional Approval

The proposed system is designed primarily as a software-based decision-support platform for electronic device assessment and repair recommendation.

The system should avoid collecting unnecessary personal information from users. Device-related information should only be collected for assessment and recommendation purposes.

The platform is also designed to support environmentally responsible decisions by encouraging repair, reuse, refurbishment, recovery, and recycling before disposal.

If the university requires an ethics declaration or institutional approval for the project, the approved documentation should be attached here.

Institutional Approval Status: [Insert actual status/reference if applicable]

Ethical Considerations:

- Avoid unnecessary collection of personal information.
- Protect stored user and device information.
- Clearly communicate that recommendations are decision-support outputs.
- Avoid presenting automated predictions as guaranteed technical diagnoses.
- Encourage professional inspection for complex or safety-critical faults.
- Promote responsible repair and e-waste management practices.

Appendix G – Project Meeting / Progress Records

This appendix contains evidence of the team's project planning, development, testing, and review activities.

The records may include:

- Initial project discussion
- Problem identification
- Literature review discussion
- Requirement analysis
- System architecture discussion
- Module development
- Database development
- Interface development
- Testing and debugging
- Faculty review

- Final demonstration preparation

Each meeting record should preferably contain:

Date: [Insert date]

Members Present: [Insert team members]

Topic Discussed: [Insert topic]

Decision Taken: [Insert decision]

Tasks Assigned: [Insert tasks]

Next Review Date: [Insert date]

Screenshots of online meetings, photographs of team discussions, progress reports, or faculty review records can be attached where appropriate.

Appendix H – User / Industry Feedback

This appendix contains feedback related to the usability and practical usefulness of the Unified Circular Economy Platform.

Feedback may be collected from students, general users, repair technicians, or other relevant reviewers.

The feedback should focus on:

- Ease of entering device information
- Ease of describing symptoms
- Clarity of fault assessment
- Usefulness of repair recommendations
- Understanding of repairability results
- Overall interface usability
- Suggestions for improving the platform

If formal user or industry feedback was not collected, the following statement may be included:

“Not Applicable – Formal user/industry feedback was not collected during the current implementation.”

If feedback was collected, attach the completed questionnaire, response screenshots, or feedback forms in this appendix.

Appendix I – Publications / Patents / Competitions / Product Outcomes

This appendix contains evidence of external outcomes related to the project, where applicable.

Possible outcomes include:

- Research paper submission
- Conference presentation

- Patent application
- Hackathon participation
- Project competition
- University-level project exhibition
- Awards or certificates
- Product demonstration
- Prototype demonstration
- Other recognized project outcomes

If no external outcome has been obtained, the following statement may be included:

“Not Applicable – No publication, patent, competition award, or external product outcome was obtained during the current project period.”

If an outcome is obtained later, the corresponding certificate, paper acceptance letter, patent information, or competition evidence should be attached here.

Appendix J – Individual Contribution Evidence

This appendix provides evidence of the individual contributions made by each team member. Since the project is a team project, individual contribution should be clearly documented.

J.1 Student 1 – Device Assessment & Machine Learning

The first student contributed primarily to the Device Assessment & Repair Recommendation module.

Major contributions included:

- Developed the device assessment logic.
- Worked on symptom preprocessing.
- Implemented machine learning-based fault assessment.
- Worked on similarity-based comparison.
- Contributed to repairability assessment.
- Developed or integrated repair recommendation logic.
- Participated in testing and debugging.

Evidence to Attach:

- Source-code screenshots
- Model/assessment screenshots
- Repository commit history
- Testing records
- Module output screenshots

J.2 Student 2 – Frontend & User Interface

The second student contributed primarily to the user interface and frontend development of the platform.

Major contributions included:

- Designed the user interface.
- Developed device information input screens.
- Developed symptom input components.
- Integrated assessment results into the interface.
- Designed the repair recommendation display.
- Improved user interaction and usability.
- Participated in system testing.

Evidence to Attach:

- UI screenshots
- Design files
- Frontend source-code screenshots
- Repository commit history
- Testing screenshots

J.3 Student 3 – Database, Backend Integration & Testing

The third student contributed primarily to database management, backend integration, and system testing.

Major contributions included:

- Designed the database structure.
- Created and managed relevant database tables.
- Worked on backend integration.
- Assisted in storing assessment information.
- Prepared test cases.
- Performed system-level testing.
- Supported debugging and final integration.

Evidence to Attach:

- Database schema screenshots
- Database table screenshots
- Backend-code screenshots
- Test-case records

- Repository commit history

J.4 Team-Level Contribution

All team members jointly contributed to the overall development of the project.

The common contributions included:

- Problem identification
- Requirement analysis
- Literature review
- System architecture
- Module planning
- Project discussions
- Integration of individual modules
- Testing and validation
- Documentation
- Final presentation
- Project demonstration

Evidence of team-level contribution may include:

- Meeting photographs
- Progress reports
- Faculty review records
- Presentation screenshots
- Final system screenshots
- Repository activity
- Testing records

The individual contribution evidence demonstrates that each team member participated in the engineering development, implementation, testing, documentation, and final presentation of the Unified Circular Economy Platform for E-Waste Recovery and Product Repair Incentivization.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)